

# Hichem Rami AIT EL HARA

PhD in Computer Science · R&D Engineer · Expert in Formal Methods & SMT Solving

📍 Palaiseau, France    ✉ [hra687261@gmail.com](mailto:hra687261@gmail.com)    🌐 [hra687261.github.io](https://hra687261.github.io)    🗣 [hra687261](#)    📄 [hra687261](#)  
 🆔 0000-0001-7909-0413    🎓 [Google Scholar](#)

## SUMMARY

Formal methods researcher and engineer, with 5 years of experience working at a French R&D consultancy, conducting studies, implementing prototypes and delivering solutions to industrial and academic clients and partners. Author and co-author of multiple publications at venues including PLDI, TACAS, and LPAR, and contributor to multiple open-source projects.

## SKILLS

**Formal Methods & Verification:** SMT Solving, Constraint Programming, Deductive Verification, Symbolic Execution, Code Transformation, Interactive Theorem Proving, Abstract Interpretation, Functional Programming, Data Structure Design and Implementation

**Programming Languages:** OCaml (Expert); Python, Lean, Rust (Novice); C, Rocq, Java, Scala (From school)

**Development Tools:** Git, VS Code, Unix, LaTeX, Typst (Experienced); Nix (Novice)

**Languages:** English (Fluent), French (Fluent), Kabyle (Native), Arabic (Fluent), Spanish (Basic)

## EXPERIENCE

**OCamlPro** [↗](#), R&D Engineer

Sept 2021 – present

- Conducted studies, implemented prototypes and delivered solutions for industrial and academic clients and partners
- Contributed to multiple open-source projects ([Alt-Ergo](#) [↗](#), [Owi](#) [↗](#), [Colibri2](#) [↗](#), [Smtml](#) [↗](#), [Dolmen](#) [↗](#), ...)
- Managed relationships with clients and prospects, organized and led technical meetings, and secured funding for R&D projects

**Université Paris-Saclay** [↗](#), **CEA List** [↗](#), PhD Student (CIFRE PhD, jointly with [OCamlPro](#) [↗](#))

Sept 2022 – Oct 2025

- Designed an SMT theory of sequences tailored for the verification of Ada programs
- Authored and co-authored multiple research papers (see Publications)
- Implemented multiple calculi to reason over SMT problems using (n-indexed) sequences and a difference logic engine
- Developed the labeled union-find data structure and used it both to reason over SMT sequences and to implement a new interval domain for real and integer arithmetic based on it

**OCamlPro** [↗](#), R&D Intern

Mar 2021 – Sept 2021

- Developed [Alt-Ergo-Fuzz](#) [↗](#), a fuzzer for the [Alt-Ergo](#) [↗](#) SMT solver, published at JFLA 2022
- Implemented context reinitialization in [Alt-Ergo](#) [↗](#), making it possible to reuse one running instance for multiple problems
- Found both runtime and subtle soundness bugs, most of which have been fixed since

## EDUCATION

**PhD** **Université Paris-Saclay** [↗](#), **CEA List** [↗](#), Computer Science

Sept 2022 – Oct 2025

- Thesis: *Theory of Sequences Tailored for Program Verification* [↗](#) (CIFRE PhD, jointly with [OCamlPro](#) [↗](#))
- [Defended on 15 October 2025](#) [↗](#)

|            |                                                                                                                                                                                                                                                                                                                                                                                     |           |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>MSc</b> | <b>Sorbonne University</b> <a href="#">↗</a> , Computer Science                                                                                                                                                                                                                                                                                                                     | Sept 2021 |
|            | <ul style="list-style-type: none"> <li>• <a href="#">Software Science and Technology (STL — Science et Technologie du Logiciel)</a> <a href="#">↗</a>, R&amp;D track focused on Software safety and reliability</li> <li>• Main subjects: Algorithmics, Formal Methods, Type Checking, Static Analysis, Program Verification, Operational Semantics and Process Algebras</li> </ul> |           |
| <b>BSc</b> | <b>Sorbonne University</b> <a href="#">↗</a> , Computer Science                                                                                                                                                                                                                                                                                                                     | Sept 2019 |

## PUBLICATIONS

---

|                                                                                                                                                                                                                                                                                                                                                                |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>colibri2: A CP solver for SMT problems</b>                                                                                                                                                                                                                                                                                                                  | June 2026 |
| François Bobot, <b>Hichem Rami Ait-El-Hara</b> , Bruno Marre<br><a href="https://cea.hal.science/cea-05644448">cea.hal.science/cea-05644448</a> <a href="#">↗</a> (Preprint)                                                                                                                                                                                   |           |
| <b>Smt.ml: A Multi-Backend Frontend for SMT Solvers in OCaml</b>                                                                                                                                                                                                                                                                                               | Apr 2026  |
| João Madeira Pereira, Filipe Marques, Pedro Adão, <b>Hichem Rami Ait-El-Hara</b> , Léo Andrès, Arthur Carcano, Pierre Chambart, Petar Maksimović, Nuno Santos, José Fragoso Santos<br><a href="https://link.springer.com/chapter/10.1007/978-3-032-22752-2_2">link.springer.com/chapter/10.1007/978-3-032-22752-2_2</a> <a href="#">↗</a> (TACAS @ ETAPS 2026) |           |
| <b>Theory of Sequences Tailored for Program Verification</b>                                                                                                                                                                                                                                                                                                   | Nov 2025  |
| <b>Hichem Rami Ait-El-Hara</b><br><a href="https://theses.fr/2025UPASG067">theses.fr/2025UPASG067</a> <a href="#">↗</a> (PhD Dissertation)                                                                                                                                                                                                                     |           |
| <b>Reasoning over n-Indexed Sequences in SMT</b>                                                                                                                                                                                                                                                                                                               | Aug 2025  |
| <b>Hichem Rami Ait-El-Hara</b> , François Bobot, Guillaume Bury<br><a href="https://link.springer.com/article/10.1007/s00236-025-00496-w">link.springer.com/article/10.1007/s00236-025-00496-w</a> <a href="#">↗</a> (Acta Informatica — Selected Extended Papers of SMT 2024 and Related Papers)                                                              |           |
| <b>Constraint Propagation for Bit-Vectors in Alt-Ergo</b>                                                                                                                                                                                                                                                                                                      | Aug 2025  |
| <b>Hichem Rami Ait-El-Hara</b> , Guillaume Bury, Basile Clément, Pierre Villemot<br><a href="https://eur-ws.org/Vol-4008">eur-ws.org/Vol-4008</a> <a href="#">↗</a> (SMT 2025: 23rd International Workshop on Satisfiability Modulo Theories)                                                                                                                  |           |
| <b>Relational Abstractions Based on Labeled Union-Find</b>                                                                                                                                                                                                                                                                                                     | June 2025 |
| Dorian Lesbre, Matthieu Lemerre, <b>Hichem Rami Ait-El-Hara</b> , François Bobot<br><a href="https://dl.acm.org/doi/10.1145/3729298">dl.acm.org/doi/10.1145/3729298</a> <a href="#">↗</a> (PLDI 2025: 46th ACM SIGPLAN Conference on Programming Language Design and Implementation)                                                                           |           |
| <b>An SMT Theory for n-Indexed Sequences</b>                                                                                                                                                                                                                                                                                                                   | July 2024 |
| Extended abstract; see the full paper above (Reasoning over n-Indexed Sequences in SMT, Acta Informatica)<br><b>Hichem Rami Ait-El-Hara</b> , François Bobot, Guillaume Bury<br><a href="https://eur-ws.org/Vol-3725">eur-ws.org/Vol-3725</a> <a href="#">↗</a> (SMT 2024: 22nd International Workshop on Satisfiability Modulo Theories)                      |           |
| <b>On SMT Theory Design: The Case of Sequences</b>                                                                                                                                                                                                                                                                                                             | May 2024  |
| <b>Hichem Rami Ait-El-Hara</b> , François Bobot, Guillaume Bury<br><a href="https://easychair.org/publications/paper/qdvj">easychair.org/publications/paper/qdvj</a> <a href="#">↗</a> (LPAR 2024: 25th Conference On Logic For Programming, Artificial Intelligence and Reasoning)                                                                            |           |
| <b>Alt-Ergo-Fuzz: A fuzzer for the Alt-Ergo SMT solver</b>                                                                                                                                                                                                                                                                                                     | June 2022 |
| <b>Hichem Rami Ait-El-Hara</b> , Guillaume Bury, Steven de Oliveira<br><a href="https://hal.inria.fr/hal-03626861">hal.inria.fr/hal-03626861</a> <a href="#">↗</a> (JFLA 2022: 33èmes Journées Francophones des Langues Applicatifs)                                                                                                                           |           |

## TALKS & SERVICE

---

See [Talks](#) [↗](#) and [Service](#) [↗](#) for a full list of invited talks, presentations, and reviewing/committee activities.